

Planning I — Classical (STRIPS)

State = set of ground atoms (**closed-world**: anything not listed is false). **Action** = (*Precond*, *Add*, *Del*); applicable if $\text{Precond} \subseteq \text{state}$; result = $(\text{state} \setminus \text{Del}) \cup \text{Add}$.

Three classic problems:

- **Frame problem:** how to specify what *stays the same*. STRIPS solves by *convention*: only listed Add/Del change; everything else persists.
- **Ramification problem:** indirect effects (e.g. if you move a briefcase, its contents move too) — not handled by STRIPS.
- **Qualification problem:** infinitely many possible preconditions (battery, weather, etc.) — pragmatic cutoff.

Search direction: *forward* (progression from init) or *backward* (regression from goal). Backward = relevant-state search.

Domain-independent heuristics: *ignore-preconditions* (drop preconds, count actions), *ignore-delete-list* (monotone $\Rightarrow$ trivial solve), $h_{\max}$, h_{add} , h_{FF} (FastForward).

Planning II — Probabilistic (MDP)

MDP tuple = (S, A, p, r) : states, actions, transition prob. $p(s'|s, a)$, reward $r(s, a)$. γ = **discount factor (NOT part of the tuple!)** — separate hyperparam, $0 \leq \gamma < 1$.

Policy $\pi : S \rightarrow A$ (det.) or $S \rightarrow \Delta(A)$ (stoch.). **Value function:**

$$V^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, \pi \right].$$

Bellman optimality:

$$V^*(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|s, a) V^*(s') \right].$$

Value iteration: iterate $V_{k+1}(s) = \max_a [\dots]$ until convergence ($V_k \rightarrow V^*$ as $k \rightarrow \infty$). **Policy iteration:** alternate *evaluation* (solve V^π) and *improvement* ($\pi'(s) \leftarrow \arg \max_a [\dots]$) until π stable. Finite iterations $\Rightarrow$ optimal.

POMDP: partial observability $\Rightarrow$ belief state over S (much harder).

Value-iteration: $V_{k+1} \leftarrow \max_a [r + \gamma \sum p V_k]$ until convergence; extract $\pi^* = \arg \max$. **Policy iteration:** alternate eval (V^π) and improve (π'). Both converge in finite time; PI does fewer, more expensive steps.

Q-function: $Q^\pi(s, a)$ = expected return from (s, a) following π . Relation: $V^\pi(s) = \sum_a \pi(a|s) Q^\pi(s, a)$; $V^*(s) = \max_a Q^*(s, a)$. Bellman: $Q^*(s, a) = r + \gamma \sum_{s'} p(s'|s, a) \max_{a'} Q^*(s', a')$. Foundation of **RL** (model-free: Q-learning, DQN; policy-gradient: REINFORCE, PPO).

Bridge to LLMs: modern reasoning LLMs (o1, DeepSeek-R1, Claude extended thinking) are trained via **RL on CoT traces** (RLHF + RLAI + outcome-reward RL). MDP = the formal substrate. *Reward shaping* matters: reward correctness of final answer + intermediate-step coherence.

Knowledge Representation I — Ontologies & DL

Goal: explicit, machine-readable symbolic knowledge. **Ontology** = formal vocabulary (concepts, roles, individuals).

Description Logic (ALC): atomic concepts A, B ; roles R ; constructors $\sqcap$ (and), $\sqcup$ (or), $\neg$ (not), $\exists R.C$ (some), $\forall R.C$ (all). Decidable. E.g. $\text{Father} \sqsubseteq \text{Male} \sqcap \exists \text{hasChild}. \text{Person}$.

TBox (terminological): concept axioms. **ABox** (assertional): instances ($\text{John} : \text{Father}$, $(\text{John}, \text{Mary}) : \text{hasChild}$).

Reasoning tasks: subsumption ($A \sqsubseteq B?$), satisfiability, classification (full taxonomy), instance checking.

OWL: web ontology language built on RDF. *Lite* (simple, decidable, fast); *DL* ($\equiv \text{SHOIN}(D)$, decidable); *Full* (RDF-compatible, undecidable). Tradeoff: **expressivity vs. decidability**.

Knowledge Representation II — RDF, SPARQL, OWA

RDF triple = (subject, predicate, object) $\rightarrow$ a graph. **RDFS** adds schema: `rdfs:Class`, `subClassOf`, `domain`, `range`.

SPARQL = SQL-like for RDF. Pattern matching with variables (`?x`); e.g. `SELECT ?p WHERE {?p :hasChild ?c}`.

Open vs Closed World Assumption (key!):

- **CWA** (databases, Prolog): *absence = false*. “Not in DB $\Rightarrow$ doesn’t exist.” Enables *negation as failure*.
- **OWA** (RDF, OWL): *absence = unknown*. “Not asserted $\Rightarrow$ we just don’t know.” Suits the web (incomplete info).

Unique Name Assumption (UNA): different names = different individuals. Standard in DB; *not* in OWL (must declare `owl:differentFrom`).

	CWA	OWA
absent fact	false	unknown
negation	as failure	classical
typical use	DB, Prolog	Web, RDF, OWL
implies UNA	yes	no

Reasoning regimes: CWA + UNA $\rightarrow$ *databases* (closed, definite). OWA + no-UNA $\rightarrow$ *open web* (incomplete, undefined identities). Mixing causes the classic “why doesn’t my query return the answer I expect?” bugs.

ALC constructors: atomic A, B ; role R ; $\sqcap$ and; $\sqcup$ or; $\neg$ not; $\exists R.C$ some; $\forall R.C$ all. **SHOIN(D)** (OWL DL) = ALC + transitive (S), hierarchy (H), nominals (O), inverse (I), numbers (N), datatypes (D).

Example — TBox: $\text{Father} \sqsubseteq \text{Male} \sqcap \exists \text{hasChild}. \text{Person}$; $\text{Parent} \sqsubseteq \exists \text{hasChild}. \text{Person}$. ABox: $\text{John} : \text{Male}$, $(\text{John}, \text{Mary}) : \text{hasChild}$, $\text{Mary} : \text{Person}$. Inferred: $\text{John} : \text{Father}$ (instance), $\text{Father} \sqsubseteq \text{Parent}$ (subsumption).

SPARQL example: `SELECT ?p WHERE {?p :hasChild ?c . ?c :age ?a . FILTER(?a < 18)}` $\rightarrow$ all parents with a minor child. Standard pattern: **triple-pattern + FILTER + projection**.

Standard reasoning tasks (DL):

- *Consistency* — is the KB satisfiable (no contradiction)?
- *Subsumption* — is $A \sqsubseteq B$? (basis for classification).
- *Classification* — compute the full concept hierarchy.
- *Instance check* — is $a : C$?

All reducible to (un)satisfiability $\Rightarrow$ **tableau algorithms**.

Complexity (rough): ALC subsumption = **ExpTime-complete**. OWL Lite $\leq$ **ExpTime**, OWL DL ($\text{SHOIN}(D)$) = **N2ExpTime**; OWL Full = **undecidable**. $\Rightarrow$ pay for what you use.

Lecture bridge: CSP (§2) and KR/DL both use *constraints* — CSP = implicit, finite-domain relations; KR = symbolic axioms with inference. Both contrast with *learned* latent representations (NN/transformers): explicit & provable vs. implicit & approximate $\Rightarrow$ **neuro-symbolic hybrids**.

Vagueness & Uncertainty

Vagueness ≠ Uncertainty:

- **Vagueness:** the predicate itself has *gradual truth* (“tall”, “warm”). → **Fuzzy logic**.
- **Uncertainty:** predicate is crisp but you’re not sure (“it rains tomorrow”). → Probability / Dempster-Shafer.

Fuzzy logic: membership $\mu_A(x) \in [0, 1]$ (not a probability!). Connectives via **T-norm** (AND, e.g. min) and **S-norm** (OR, e.g. max); negation $1 - \mu$. Fuzzy implication $\mu_{A \rightarrow B} = \min(1, 1 - \mu_A + \mu_B)$ (Łukasiewicz).

Probability: crisp facts + degree of belief in $[0, 1]$, sums to 1, Bayes’ rule for updating. Cannot represent *ignorance* (no info = uniform = a strong claim).

Dempster-Shafer: mass function $m : 2^\Omega \rightarrow [0, 1]$ over subsets, $\sum m(A) = 1$. Two measures:

- **Belief** $BEL(A) = \sum_{B \subseteq A} m(B)$ — evidence for.
- **Plausibility** $PL(A) = \sum_{B \cap A \neq \emptyset} m(B)$ — not refuted.

$PL(A) - BEL(A) = \text{ignorance}$. Generalises probability (m on singletons only $\Rightarrow$ probability). Dempster’s rule combines independent evidence.

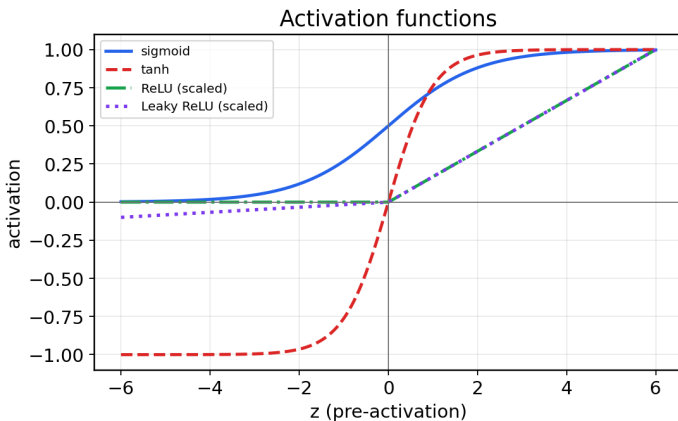
Neural Networks

Perceptron: $y = \text{sign}(w^\top x + b)$. Linear classifier; converges *iff* data linearly separable (XOR fails $\Rightarrow$ MLP needed).

MLP (Multi-Layer Perceptron): stacked layers $h^{(\ell+1)} = \sigma(W^{(\ell)} h^{(\ell)} + b^{(\ell)})$. **Universal approximation theorem:** 1 hidden layer of finite width can approximate any continuous function (in principle — needs exponential width).

Activations:

- **Sigmoid** $\sigma(z) = 1/(1 + e^{-z}) \in (0, 1)$ — saturates $\Rightarrow$ *vanishing gradient*.
- **tanh** $\in (-1, 1)$ — zero-centred, still saturates.
- **ReLU** $\max(0, z)$ — sparse, fast, no upper saturation. *Dying ReLU*: stuck at 0.
- **softmax** for multi-class output ($\sum = 1$).



Loss: MSE (regression), cross-entropy $-\sum y_i \log \hat{y}_i$ (classification).

Backpropagation: chain rule, propagate $\partial L / \partial w$ backward layer-by-layer. **SGD** updates $w \leftarrow w - \eta \nabla L$; **mini-batch**, *momentum*, **Adam** (adaptive learning rate).

Deep nets: vanishing/exploding gradients (long chain of sigmoids $\rightarrow 0$); *fixes*: ReLU, residual connections, batch norm, careful init (Xavier/He).

Regularisation: dropout (randomly zero units), weight decay (L2), early stopping, data augmentation. **Optimisers:** SGD, *momentum*, *RMSProp*, **Adam** (default for deep nets). **Perceptron limit:** linearly separable only (XOR fails) $\Rightarrow$ MLP needed.

Deep Learning & Transformers

CNN: *convolutional* layer = shared-weight local filters ($\Rightarrow$ translation invariance, far fewer params). **Pooling** (max/avg) down-

samples $\Rightarrow$ larger receptive field. Hierarchy: edges $\rightarrow$ shapes $\rightarrow$ objects. **ResNet:** *skip connections* $h + f(h)$ solve vanishing gradient $\Rightarrow$ very deep nets trainable.

Transformer block = self-attention + FFN + residual + layer-norm. Stack N blocks.

Self-attention per head:

$$\text{Attn}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$$

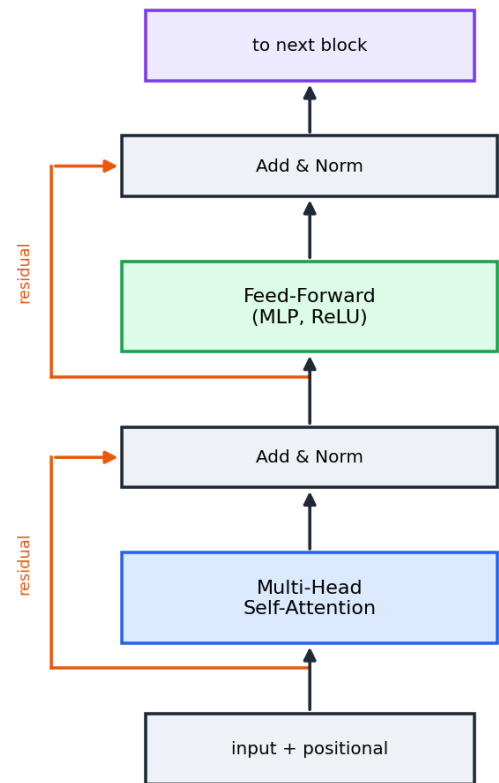
$Q = XW_Q$, $K = XW_K$, $V = XW_V$, where X = token embedding + position. **Multi-head** = h heads in parallel, concatenated. Cost: $O(n^2 d)$ in sequence length n .

Positional encoding: *absolute* (sinusoidal/learned, added or concatenated), *relative*, or **RoPE** (rotation of Q/K by angle $\propto$ position $\Rightarrow$ relative distance in the dot product). Required because attention is permutation-invariant.

Encoder (BERT): bidirectional attention; pretraining via Masked-LM. **Decoder (GPT):** *causal mask*; next-token prediction. Encoder-decoder (T5, original): seq2seq.

Pretraining $\rightarrow$ **fine-tuning** paradigm: huge unsupervised pre-train, small supervised fine-tune. **Scaling laws** (Kaplan, Chin-chilla): loss decays as power-law in params/data/compute; Chin-chilla $\Rightarrow$ *compute-optimal* tokens $\approx 20 \times$ params.

One transformer block
(decoder adds a masked self-attention + cross-attention)



Limitations: hallucinations, no reliable reasoning in a single pass (see exam paper!), context length quadratic in attention, compute-hungry, opaque (mechanistic interp).

Bridges to other topics: (i) MDP planning is the foundation of RL $\rightarrow$ used to train modern reasoning LLMs (RLHF, RLAI, RL-from-CoT). (ii) Transformers *learn* representations but *can't* reliably do classical search (paper) $\Rightarrow$ neuro-symbolic hybrids couple LLMs with symbolic solvers (KR, planning).

Rapid defs: γ part of MDP? no, separate. *TBox/ABox?* terminology / instances. *Fuzzy vs prob. vs DS?* vagueness / uncertainty / uncertainty + ignorance. *ReLU vs sigmoid?* no saturation $\Rightarrow$ no vanishing gradient. *Why position encoding?* attention is permutation-invariant.